Question Paper

Exam Date & Time: 20-Nov-2023 (01:15 PM - 04:30 PM)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination Nov 2023
Seventh Semester of B.Tech. (CS)

Design of Language Processor [CS450]

Marks: 70 **Duration: 195 mins.**

SECTION - I

Answer all the questions. Section Duration: 40 mins

1 Which of the following is responsible for generating an intermediate representation of the source program? (1)

Syntax analyzer	Semantic analyzer	Code generator	Intermediate code generator

2 Which phase of a language processor is responsible for syntax analysis and error detection? (1)

Lexical analysis	Semantic analysis	Code optimization	Syntax analysis

3 What is the primary role of the semantic analyzer in a language processor? (1)

Error detection	Symbol table generation	Tokenization	Type checking

4 Which phase of the language processor is responsible for converting the intermediate code into the target machine code? (1)

Code optimization	Code generator	Lexical analysis	Syntax analysis

5 What is the primary purpose of a symbol table in a language processor? (1)

Error detection	Intermediate code generation	Memory management	Symbol management

6 Which phase of the language processor performs code transformations to improve its quality without changing its functionality? (1)

Lexical analysis	Syntax analysis	Code optimization	Semantic analysis

7 What does the syntax analyzer ensure in a language processor? (1)

Consistency in variable naming	Code efficiency	Compliance with language grammar rules	Appropriate use of comments

8 Which of the following generates a syntax tree or an abstract syntax tree (AST) in a language processor? (1)

Code generator | Parser | Lexical analyzer | Semantic analyzer

9 | | | Which phase of a language processor involves the creation and management of temporary variables and addresses? (1)

Code generation | Lexical analysis | Semantic analysis | Code optimization

10 | | | What is the primary function of the intermediate code in a language processor? (1)

Direct execution on the CPU | Code optimization | Providing a link between the source code and the target code | Error detection

11 | | | What is the role of the error recovery module in a language processor? (1)

Ignoring errors | Reporting errors | Handling errors gracefully | Avoiding errors

12 | | | Which phase of a language processor is responsible for identifying and resolving references to variables and their corresponding memory locations? (1)

Semantic analysis | Code optimization | Code generation | Syntax analysis

13 | | | Match all items in Group 1 with correct options from those given in Group 2. (1)

Group 1 Group 2
P. Regular expression 1. Syntax analysis
Q. Pushdown automata 2. Code generation
R. Dataflow analysis 3. Lexical analysis
S. Register allocation 4. Code optimization

P-4. Q-1, R-2, S-3 | P-3, Q-1, R-4, S-2 | P-3, Q-4, R-1, S-2 | P-2, Q-1, R-4, S-3

14 | | | An LALR(1) parser for a grammar G can have shift-reduce (S-R) conflicts if and only if (1)

the SLR(1) parser for G has S-R conflicts | the LR(1) parser for G has S-R conflicts | the LR(0) parser for G has S-R conflicts | the LALR(1) parser for G has reduce-reduce conflicts

15 | | | Assume that the SLR parser for a grammar G has n1 states and the LALR parser for G has n2 states. The relationship between n1 and n2 is: (1)

n1 is necessarily less than n2 | n1 is necessarily equal to n2 | n1 is necessarily greater than n2 | none of these

16 | | | In a bottom-up evaluation of a syntax directed definition, inherited attributes can (1)

always be evaluated | be evaluated only if the definition is L-attributed | be evaluated only if the definition has synthesized attributes | never be evaluated

17 | | | The number of tokens in the following C statement is printf("i=%d, &i=%x", i,&i); (1)

13 | 11 | 10 | 9

18 | Which one from the following is false? (1)

- LALR parser is Bottom - Up parser
- A parsing algorithm which performs a left to right scanning and a right most deviation is RL (1)
- LR parser is Bottom - Up parser.
- In LL(1), the 1 indicates that there is a one - symbol look - ahead.

19 | In the context of abstract-syntax-tree (AST) and control-flow-graph (CFG), which one of the following is True? (1)

- In both AST and CFG, let node N2 be the successor of node N1. In the input program, the code corresponding to N2 is present after the code corresponding to N1
- For any input program, neither AST nor CFG will contain a cycle
- The maximum number of successors of a node in an AST and a CFG depends on the input program
- Each node in AST and CFG corresponds to at most one statement in the input program

20 | Match the following: (1)

List-I List-II
A. Lexical analysis 1. Graph coloring
B. Parsing 2. DFA minimization
C. Register allocation 3. Post-order traversal
D. Expression evaluation 4. Production tree

Codes:
A B C D
(a) 2 3 1 4
(b) 2 1 4 3
(c) 2 4 1 3
(d) 2 3 4 1

- a
- b
- c
- d

SECTION - II

Answer 5 out of 7 questions.

1 | Define symbol table and outline the purposes for which a compiler utilizes it. (4)

2 | Determine the First and Follow sets for the provided grammar. (4)
$S \rightarrow AaB \mid bC$
$A \rightarrow cD \mid \varepsilon$
$B \rightarrow e \mid \varepsilon$
$C \rightarrow f \mid \varepsilon$
$D \rightarrow g \mid \varepsilon$

3 | Explain the various issues in the design of a code generator. (4)

4 | Define lexical analysis and enumerate the tasks carried out by a lexical analyzer. (4)

5 | Construction of Non-recursive Predictive Parsing Table. (4)

6 |

Describe the process of handling syntax errors. (4)

7 Describe Basic-Block Scheduling. (4)

SECTION - III

Answer 6 out of 8 questions.

1 Give the rule to remove left recursive grammar. And Eliminate left recursion from following grammar. (5)
$S \rightarrow Aa \mid b$
$A \rightarrow Ac \mid Sd \mid f$

2 Provide an explanation of synthesized attributes, supported by an illustrative example. (5)

3 Construct CLR parsing table for following grammar. (5)
S -> aSA | €
A -> bS | c

4 Discuss about S-attributed and L-attributed SDT with example. (5)

5 Show the following grammar is LR(1) but not LALR(1). (5)
S->Aa │bAc │Bc│bBa
A->d
B->d

6 Provide an explanation of the input, output, and actions carried out by each phase of the compiler, accompanied by an example. (5)

7 Define peephole optimization and illustrate it with an example. (5)

8 Describe three address code generators with example. (5)

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